



Sri Lanka Institute of Information Technology
**Health Guard Medical Center
Management System**

Sprint 0

Information Systems Project Management 2026
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Abstract

The Health Guard Medical Center Management System is a web-based application developed to improve the way daily activities are managed within a medical center. Currently, many activities such as patient appointments, clinical records, medicine inventory, staff information, billing, and payments are handled manually or through separate records. This can result in repeated data entry, delays in communication, difficulty in tracking information, medicine stock issues, payment errors, and limited visibility for management.

The proposed system brings these major activities together into one centralized and role-based web application. The system is organized into four main areas: User and Staff Management, Appointment and Clinical Workflow, Inventory and Procurement, and Billing and Payment Management. It allows patients to manage appointments, while doctors can record consultations, prescriptions, and laboratory requests. Nurses can manage patient check-ins and vital information, pharmacists can maintain medicine and supplier records, and cashiers can manage invoices, payments, and receipts. In addition, managers can monitor staff attendance, inventory information, financial reports, and attendance-based payroll.

The main goal of the project is to connect these activities and provide accurate, accessible, and traceable information through a single system. By reducing manual work and improving coordination between different roles, the system is expected to support smoother medical-center operations, improve patient service, strengthen inventory control, and make financial and workforce management more efficient. The project will be developed following an iterative approach over the planned 13-week period, covering system design, development, testing, bug fixing, and final completion.

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Assessment Criteria

Criteria	Marks allocated	Marks given
Problem Specification	5%	
Business Idea	5%	
Solution Outline	10%	
Key Benefits	10%	
Main Objectives	5%	
Sub Objectives	5%	
Project Plan	10%	
Product Vision statement	5%	
Epics	5%	
User Stories	10%	
Initial Product Backlog	5%	
Definition of Done	5%	
Hardware & Software requirements	5%	
Budget	5%	
References	5%	
Formatting	5%	
Total Marks	100%	

Team Members:

Student ID	Student Name	Role (Scrum Master/Product Owner/Team Member)
IT24100799	Gawrawa G H Y	Scrum Master
IT24101266	Perera M U E	Team Member
IT24100264	Bandara N W C D	Product owner
IT24100344	Gunasekara W L L	Team Member

1. Introduction

1.1 Problem Specification

Health Guard Medical Center currently manages many daily tasks manually, such as appointments, prescriptions, medicine stock, staff records, and payments. Because this information is kept in different places, staff may need to enter the same details more than once, communication can be delayed, and it can be difficult to see the latest information quickly.

The main problem is that the medical center does not have one system that connects all of these activities. This can lead to longer patient waiting times, difficulties in tracking medicine stock and expiry dates, payment errors, and limited visibility for managers. The project aims to solve this by bringing the main medical center activities together in one integrated web application.

The project addresses the following current-state issues:

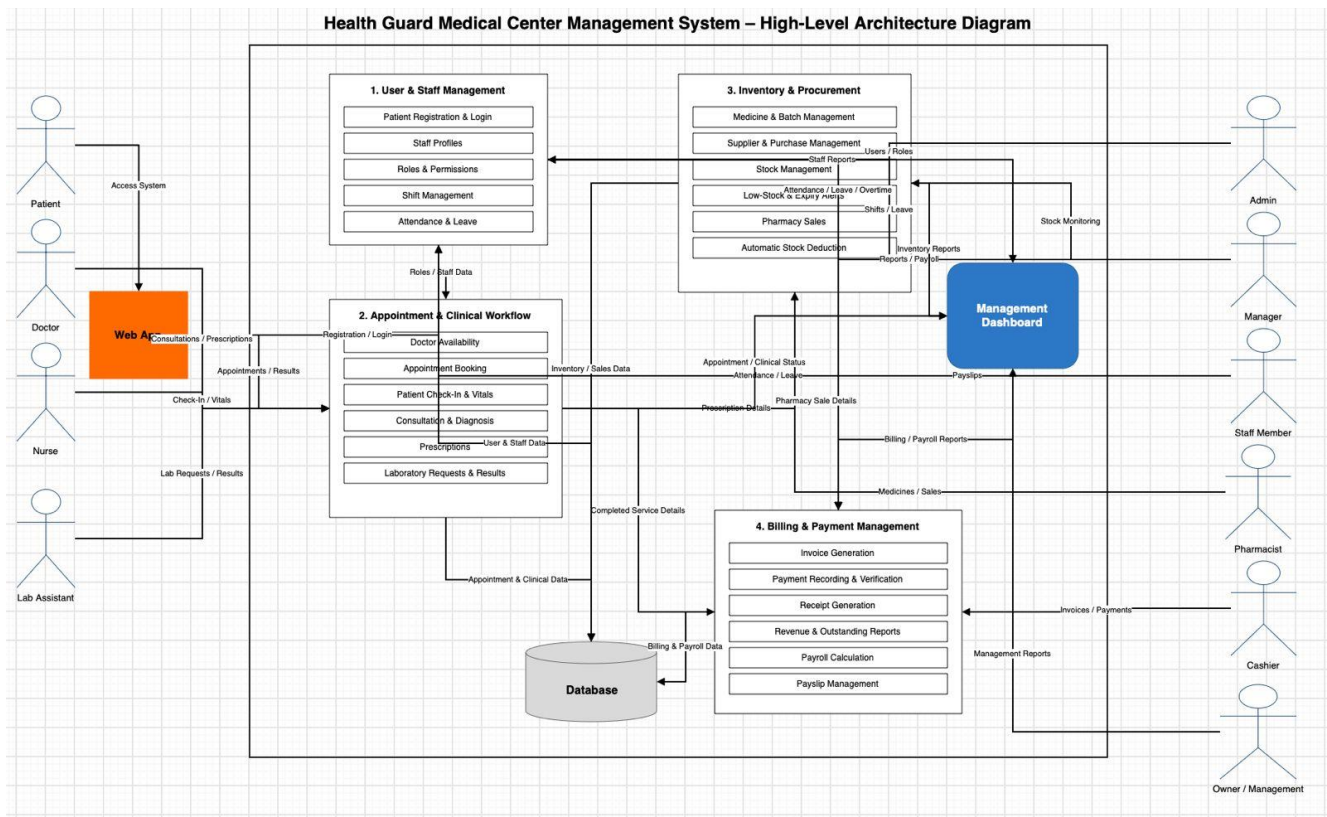
- Patient appointments, check-ins, clinical records and laboratory activities are not coordinated through one standardized workflow.
- Staff profiles, roles, shifts, attendance and leave information are maintained separately, making workforce monitoring and payroll preparation more difficult.
- Medicine, batch, supplier, stock and expiry information is not centrally connected to dispensing and pharmacy sales activities.
- Invoices, payments, receipts and outstanding balances are not maintained through one traceable billing workflow.
- Attendance information is not directly connected to payroll calculations, increasing the risk of manual calculation errors.
- Management has limited real-time visibility across patient services, staff operations, inventory status and financial information.

1.2 Business Idea

Health Guard Medical Center Management System is a web application that helps manage the main activities of the medical center in one place. It includes staff management, appointments, clinical activities, medicine inventory, billing, and payments. The system connects these activities together. Patients can book appointments, doctors can create prescriptions and lab requests, pharmacists can manage medicine stock, cashiers can handle payments, and managers can manage staff attendance and salaries. This makes the medical center easier to manage and reduces manual work.

1.3 Solution Outline

The Health Guard project will use a simple web-based structure with responsive frontend, backend/API, central database, and role-based access control. The project does not require a specific technology stack, so the team can choose or change the technologies later without affecting the main system requirements.



The application layer coordinates the four epic services and enforces business rules such as role authorization, appointment status transitions, laboratory workflow status, batch-level stock updates, invoice/payment status and attendance-based payroll calculations. The project charter explicitly excludes external laboratory systems, insurance claim integrations and payment gateway APIs from the MVP, so these integrations are not represented as required implementation dependencies.

1.4 Key Benefits

- Keep patient and staff details in one place.
- Give each user access only to the functions they need.
- Make appointment booking and patient check-in easier.
- Keep consultations, prescriptions, and lab records connected.
- Track medicine stock and expiry dates.
- Update medicine stock automatically after sales.
- Manage invoices, payments, and receipts in one system.
- Use attendance and leave records for salary calculations.
- Help managers view important reports.
- Allow the system to add more features in the future.

2.Objectives

2.1 Main Objective

To design, develop, integrate and validate within the 13-week project period a role-based web-based Medical Center Management System for Health Guard Medical Center that integrates user and staff management, appointment and clinical workflows, medicine inventory and procurement, and billing, payment and payroll activities into a single coordinated platform, with successful completion demonstrated through the integrated Minimum Viable Product (MVP), functional testing and scheduled sprint evaluations.

2.2 Sub Objectives

- To design, develop and validate the **User and Staff Management component** within the 13-week project period, supporting patient and staff account management, role-based access control, staff profiles, shift management, attendance and leave management, with completion measured through successful functional testing and acceptance of the relevant epic user stories.
- To design, develop and validate the **Appointment and Clinical Workflow component** within the 13-week project period, supporting appointment availability and booking, patient check-in, vital-sign recording, consultations, prescriptions, laboratory requests, test progress and laboratory results, with completion measured through successful functional and end-to-end workflow testing and acceptance of the relevant epic user stories.
- To design, develop and validate the **Inventory and Procurement component** within the 13-week project period, supporting medicine records, medicine batches, supplier management, stock quantities, procurement, expiry monitoring, low-stock alerts, pharmacy sales and automatic stock updates, with completion measured through successful functional testing and acceptance of the relevant epic user stories.
- To design, develop and validate the **Billing and Payment Management component** within the 13-week project period, supporting patient invoices, payment recording and verification, receipts, outstanding balances, financial reporting and attendance-based payroll processing, with completion measured through successful functional and end-to-end workflow testing and acceptance of the relevant epic user stories.

Achievement of these objectives will be evaluated through completion and acceptance of the corresponding epic user stories, successful functional and integration testing, and demonstration of the integrated Health Guard MVP during the scheduled sprint evaluations.

3. Stakeholder Identification

Stakeholder	Type (Primary / Secondary)	Interest / Concern
Owner / Medical Center Management	Primary	Organizational direction, system approval, operational visibility, financial and workforce reporting.
Admin	Primary	Accurate staff records, user accounts, roles and permissions.
Manager	Primary	Shift scheduling, leave approval, attendance monitoring, inventory alerts, reports and payroll.
Doctor	Primary	Appointment schedule, consultation records, diagnoses, prescriptions, lab requests and result review.
Nurse	Primary	Patient check-in, queue readiness and vital-sign recording.
Pharmacist	Primary	Medicine, batch, supplier and stock management; dispensing, pharmacy sales and reports.
Cashier	Primary	Patient invoices, payment recording/verification, reconciliation and receipts.
Laboratory Assistant	Primary	Laboratory request verification, test progress and result entry/upload.
Patients	Secondary	Account access, appointments, lab reports, invoices, balances and receipts.
Medicine Suppliers	Secondary	Reliable purchase/stock coordination and accurate supplier records.

4. Procedure

The project follows an iterative, sprint-based implementation approach across the 13-week project period. Sprint 0 establishes the project scope, stakeholders, product vision, epics, user stories, initial product backlog, effort estimates, quality criteria, system architecture and project plan. Development then proceeds through Sprint 1, Sprint 2 and Sprint 3. During each sprint, the four major Health Guard components are developed in parallel and progressively integrated. Functional, integration, security and usability testing are carried out throughout development, and identified defects are corrected before final completion. Sprint evaluation points are used to review progress and refine the remaining backlog.

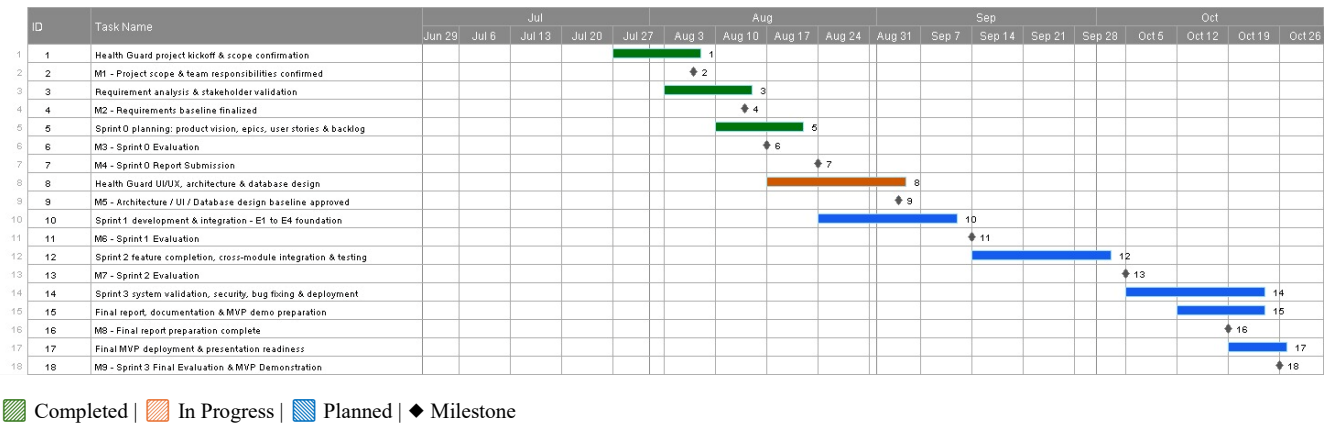


Figure 2. Gantt chart aligned with the 13-week Health Guard project timeline.

Key project milestones are:

- Week 3: Sprint 0 evaluation.
- Week 4: Sprint 0 report submission.
- Week 7: Sprint 1 evaluation.
- Week 10: Sprint 2 evaluation.
- Week 12: Final report submission preparation.
- Week 13: Sprint 3 final evaluation and integrated MVP demonstration.

5. Product Vision Statement

Vision Element	Statement
For	Health Guard Medical Center, its staff and patients,
Who	need a reliable way to coordinate workforce records, appointments, clinical activities, medicines, billing and payroll,
The product	Health Guard Medical Center Management System is an integrated role-based web application,
Provides	a centralized and traceable workflow that improves operational visibility, information accuracy, patient service coordination, inventory control and financial administration.

6. Epics

The integrated project is organized into four significant end-to-end epics so that each team member owns one major process area while all four areas remain connected through shared data and workflows. Each epic is independently owned for development and assessment, but the final MVP is validated as one integrated Health Guard system.

Epic ID	Epic	Epic Owner	Epic Description
E1	User & Staff Management	Gawrawa G H Y (IT24100799)	Manages patient accounts and medicalcenter workforce records, including staff profiles, role-based access, shifts, attendance and leave.
E2	Appointment & Clinical Workflow	Perera M U E (IT24101266)	Connects appointment booking, patient check-in, vital recording, consultations, prescriptions and laboratory-test workflows.
E3	Inventory & Procurement	Bandara N W C D (IT24100264)	Manages medicines, suppliers, batches, stock quantities, expiry monitoring, pharmacy dispensing/sales and related reports.
E4	Billing & Payment Management	Gunasekara W L L (IT24100344)	Manages patient service billing, payment records, receipts, reconciliation, financial reporting and attendance-based staff payroll.

7. User Stories

7.1 E1 - User & Staff Management

Epic ID	User Story ID	User Story	Stakeholder
E1	E1-US01	As a Patient, I want to create my own account so that I can book appointments and access medical-center services online.	Patient
E1	E1-US02	As an Admin, I want to register new staff members so that all medical-center employees can access the system according to their responsibilities.	Admin
E1	E1-US03	As an Admin, I want to update staff information so that workforce records remain accurate and current.	Admin
E1	E1-US04	As an Admin, I want to assign roles and permissions to staff members so that users can access only the functions they are authorized to use.	Admin
E1	E1-US05	As a Manager, I want to create and manage staff work shifts so that appropriate staff are available during medical-center operating hours.	Manager
E1	E1-US06	As a Staff Member, I want to record my attendance so that my work history remains accurate.	Staff Member
E1	E1-US07	As a Staff Member, I want to submit leave requests so that I can request time away from work.	Staff Member
E1	E1-US08	As a Manager, I want to review staff leave requests so that staffing levels can be properly maintained.	Manager
E1	E1-US09	As a Staff Member, I want to view my attendance records so that I can verify my working history.	Staff Member
E1	E1-US10	As a Manager, I want to view staff attendance information so that I can monitor staff availability and performance.	Manager

7.2 E2 - Appointment & Clinical Workflow

Epic ID	User Story ID	User Story	Stakeholder
E2	E2-US01	As a Patient, I want to view available appointment slots so that I can select a suitable date and time.	Patient
E2	E2-US02	As a Patient, I want to book an available appointment so that I can consult a doctor.	Patient
E2	E2-US03	As a Patient, I want to view my current and previous appointments so that I can track my medical visits.	Patient
E2	E2-US04	As a Patient, I want to cancel an appointment before the scheduled time so that the appointment slot can become available to another patient.	Patient
E2	E2-US05	As a Nurse, I want to check in arriving patients so that doctors know which patients are ready for consultation.	Nurse
E2	E2-US06	As a Nurse, I want to record patient vital information so that the doctor can review the patient's condition before consultation.	Nurse
E2	E2-US07	As a Doctor, I want to record consultation details so that the patient's medical history is maintained.	Doctor
E2	E2-US08	As a Doctor, I want to create a prescription so that the patient can obtain the required medicines.	Doctor
E2	E2-US09	As a Doctor, I want to create a laboratory test request so that the patient can complete required medical tests.	Doctor
E2	E2-US10	As a Patient, I want to receive a notification when my doctor requests a laboratory test so that I know which test needs to be completed.	Patient
E2	E2-US11	As a Lab Assistant, I want to verify the patient's laboratory request so that I can confirm the requested test before processing it.	Lab Assistant

E2	E2-US12	As a Lab Assistant, I want to update the progress of laboratory tests so that patients and doctors can know the current test status.	Lab Assistant
E2	E2-US13	As a Lab Assistant, I want to enter or upload laboratory results so that the patient and doctor can access the completed report.	Lab Assistant
E2	E2-US14	As a Patient, I want to view my completed laboratory reports so that I can access my medical test results.	Patient
E2	E2-US15	As a Doctor, I want to view laboratory results requested for my patients so that I can make further clinical decisions.	Doctor

7.3 E3 - Inventory & Procurement

Epic ID	User Story ID	User Story	Stakeholder
E3	E3-US01	As a Pharmacist, I want to add new medicines so that medicines available at the medical center are recorded in the system.	Pharmacist
E3	E3-US02	As a Pharmacist, I want to record medicine batches so that stock can be monitored according to batch and expiry information.	Pharmacist
E3	E3-US03	As a Pharmacist, I want to view all medicines and available quantities so that I can monitor current inventory.	Pharmacist

E3	E3-US04	As a Pharmacist, I want to search for medicines so that I can quickly find medicine information.	Pharmacist
E3	E3-US05	As a Pharmacist, I want to view individual medicine batches so that I can monitor quantities and expiry dates.	Pharmacist
E3	E3-US06	As a Pharmacist, I want to update medicine information so that medicine records remain accurate.	Pharmacist
E3	E3-US07	As a Pharmacist, I want to update stock quantities so that inventory levels reflect actual stock availability.	Pharmacist

E3	E3-US08	As a Pharmacist, I want to update medicine batch information so that incorrect or changed batch details can be corrected.	Pharmacist
E3	E3-US09	As a Pharmacist, I want to update medicine prices so that pharmacy sales use the correct current prices.	Pharmacist
E3	E3-US10	As a Pharmacist, I want to record supplier information so that the source of medicine purchases can be tracked.	Pharmacist
E3	E3-US11	As a Pharmacist, I want to record medicine purchases from suppliers so that new stock can be added accurately.	Pharmacist
E3	E3-US12	As a Manager, I want to receive low-stock alerts so that medicines can be reordered before they become unavailable.	Manager
E3	E3-US13	As a Manager, I want to receive medicine expiry alerts so that expired medicines and unnecessary wastage can be reduced.	Manager
E3	E3-US14	As a Pharmacist or Manager, I want to view low-stock medicines so that purchasing decisions can be made.	Pharmacist / Manager
E3	E3-US15	As a Pharmacist or Manager, I want to view medicine expiry information so that expiring medicines can be managed appropriately.	Pharmacist / Manager
E3	E3-US16	As a Pharmacist, I want to sell medicines through the system so that pharmacy transactions are properly recorded.	Pharmacist
E3	E3-US17	As a Pharmacist, I want to generate a bill for medicines sold so that the patient receives evidence of the transaction.	Pharmacist
E3	E3-US18	As a Pharmacist, I want the system to automatically reduce medicine quantities after a sale so that stock records remain accurate.	Pharmacist
E3	E3-US19	As a Pharmacist, I want to view previous medicine sales so that I can track pharmacy transactions.	Pharmacist
E3	E3-US20	As a Pharmacist or Manager, I want to view pharmacy sales reports so that medicine sales performance can be monitored.	Pharmacist / Manager

7.4 E4 - Billing & Payment Management

Epic ID	User Story ID	User Story	Stakeholder
E4	E4-US01	As a Cashier, I want to generate an invoice from finalized medical services so that the patient can pay the correct amount.	Cashier
E4	E4-US02	As a Patient, I want to view my invoices so that I know how much I need to pay.	Patient
E4	E4-US03	As a Patient, I want to view my outstanding balance so that I know whether I have unpaid medical bills.	Patient
E4	E4-US04	As a Cashier, I want to record patient payments so that invoices can be marked as paid.	Cashier
E4	E4-US05	As a Cashier, I want to verify payments so that only valid transactions are recorded.	Cashier
E4	E4-US06	As a Cashier, I want to reconcile recorded payments so that financial records match actual collected payments.	Cashier
E4	E4-US07	As a Cashier, I want the system to generate a receipt after successful payment so that the patient receives proof of payment.	Cashier
E4	E4-US08	As a Patient, I want to view or download my payment receipt so that I have proof of my payment.	Patient
E4	E4-US09	As a Manager, I want to view revenue reports so that I can	Manager
		monitor the financial performance of the medical center.	
E4	E4-US10	As a Manager, I want to view outstanding invoices so that unpaid patient balances can be monitored.	Manager
E4	E4-US11	As an Admin or Manager, I want to maintain staff salary information so that salaries can be calculated correctly.	Admin / Manager
E4	E4-US12	As a Manager, I want the system to obtain staff attendance records so that salary calculations are based on actual attendance.	Manager

E4	E4-US13	As a Manager, I want the system to calculate each employee's monthly salary so that payroll can be processed efficiently.	Manager
E4	E4-US14	As a Manager, I want to review calculated salaries before payment so that incorrect payroll calculations can be identified.	Manager
E4	E4-US15	As a Manager, I want to approve staff salary payments so that only reviewed payroll records are finalized.	Manager
E4	E4-US16	As a Manager, I want to record completed salary payments so that payroll history remains accurate.	Manager
E4	E4-US17	As a Manager, I want the system to generate a payslip so that each employee can see details of their salary calculation.	Manager
E4	E4-US18	As a Staff Member, I want to view my monthly payslip so that I can understand how my salary was calculated.	Staff Member
E4	E4-US19	As a Staff Member, I want to view my salary payment history so that I can keep track of previous salary payments.	Staff Member
E4	E4-US20	As a Manager, I want to view payroll reports so that staff salary expenses can be monitored.	Manager

8. Initial Product Backlog

The initial product backlog is organized epic-wise and prioritized according to business value, dependency and implementation readiness. Story points are used as relative effort estimates. For initial academic planning, the team uses approximately four team-hours per story point as a rough effort-conversion assumption; both story-point and hour estimates may be refined based on sprint performance and implementation feedback.

8.1 E1 - User & Staff Management

User Story ID	Remarks	Estimated story points	Estimated hours	Priority
E1-US01	Foundation patient access	3	12	High
E1-US02	Foundation staff master data	5	20	High
E1-US03	Staff record accuracy	3	12	Medium
E1-US04	Authorization and RBAC	5	20	High
E1-US05	Workforce scheduling	5	20	High
E1-US06	Attendance source for payroll	3	12	High
E1-US07	Leave workflow input	3	12	Medium
E1-US08	Staffing control	3	12	Medium
E1-US09	Employee self-service	2	8	Low
E1-US10	Management workforce visibility	3	12	Medium

8.2 E2 - Appointment & Clinical Workflow

User Story ID	Remarks	Estimated story points	Estimated hours	Priority
E2-US01	Appointment availability	3	12	High
E2-US02	Core booking transaction	5	20	High
E2-US03	Patient appointment traceability	3	12	Medium
E2-US04	Releases unused slots	3	12	Medium

E2-US05	Starts clinical visit	3	12	High
E2-US06	Pre-consultation clinical data	3	12	High
E2-US07	Core consultation record	5	20	High
E2-US08	Links clinical work to pharmacy	5	20	High
E2-US09	Starts lab workflow	5	20	High
E2-US10	Patient awareness and reference	2	8	Medium
E2-US11	Lab request control	3	12	High
E2-US12	Test status visibility	3	12	Medium
E2-US13	Clinical result record	5	20	High
E2-US14	Patient result access	3	12	Medium
E2-US15	Doctor follow-up decision support	3	12	Medium

8.3 E3 - Inventory & Procurement

User Story ID	Remarks	Estimated story points	Estimated hours	Priority
E3-US01	Medicine master data	3	12	High
E3-US02	Batch/expiry foundation	5	20	High
E3-US03	Inventory visibility	3	12	High
E3-US04	Fast retrieval	2	8	High
E3-US05	Batch-level monitoring	2	8	Medium
E3-US06	Record maintenance	3	12	Medium
E3-US07	Stock accuracy	3	12	High
E3-US08	Batch correction	3	12	Medium
E3-US09	Sales price accuracy	2	8	Medium
E3-US10	Supplier master data	3	12	High

E3-US11	Procurement stock intake	5	20	High
E3-US12	Prevents stock-outs	3	12	High
E3-US13	Reduces expiry wastage	3	12	High
E3-US14	Purchasing support	3	12	Medium
E3-US15	Expiry management	3	12	Medium
E3-US16	Core pharmacy transaction	8	32	High
E3-US17	Patient sale evidence	3	12	High
E3-US18	Critical stock automation	5	20	High
E3-US19	Transaction traceability	3	12	Medium
E3-US20	Operational reporting	5	20	Medium

8.4 E4 - Billing & Payment Management

User Story ID	Remarks	Estimated story points	Estimated hours	Priority
E4-US01	Core patient billing	5	20	High
E4-US02	Patient billing visibility	2	8	Medium
E4-US03	Unpaid balance visibility	2	8	Medium
E4-US04	Core payment transaction	5	20	High
E4-US05	Payment integrity	3	12	High
E4-US06	Financial reconciliation	5	20	Medium
E4-US07	Payment evidence	3	12	High
E4-US08	Patient self-service	2	8	Medium
E4-US09	Financial monitoring	5	20	Medium
E4-US10	Receivables monitoring	3	12	Medium
E4-US11	Payroll foundation	3	12	High

E4-US12	Epic 1 to payroll integration	5	20	High
E4-US13	Core payroll calculation	8	32	High
E4-US14	Payroll control	3	12	Medium
E4-US15	Approval governance	3	12	Medium
E4-US16	Payroll history	3	12	Medium
E4-US17	Employee payment evidence	5	20	High
E4-US18	Employee self-service	2	8	Medium
E4-US19	Payroll traceability	2	8	Low
E4-US20	Payroll reporting	5	20	Medium

9. Definition of Done (DoD)

A user story or epic is considered Done when:

- The agreed acceptance criteria and business rules for the item are implemented.
- The user interface, backend/API and required database operations are integrated and data persists correctly.
- Input validation, authentication, role-based access and relevant security controls are applied.
- Required cross-component data flows update correctly without breaking dependent workflows.
- The implementation has been peer-reviewed by at least one other team member.
- Functional tests for normal, boundary and key error scenarios have passed.
- No unresolved critical or high-severity defects remain for the completed scope.
- The feature works in the shared integration/test environment and does not break previously completed core workflows.
- Relevant technical or user documentation is updated where required.
- The story or epic is demonstrated and accepted by the Product Owner/team against its acceptance criteria.

10. Sprint 1 Preview

Sprint 1 selects at least 50% of the user stories from every epic, prioritizing foundation data, core transactions and cross-epic dependencies needed for later workflows. The selected stories will be reviewed during the Sprint 1 evaluation, and unfinished items will be returned to the backlog for reprioritization.

Epic	User Story ID	Sprint 1 Scope	Story Points	Priority
E1	E1-US01	As a Patient, I want to create my own account so that I can book appointments and access medical-center services online.	3	High
E1	E1-US02	As an Admin, I want to register new staff members so that all medical-center employees can access the system according to their responsibilities.	5	High
E1	E1-US03	As an Admin, I want to update staff information so that workforce records remain accurate and current.	3	Medium

E1	E1-US04	As an Admin, I want to assign roles and permissions to staff members so that users can access only the functions they are authorized to use.	5	High
E1	E1-US05	As a Manager, I want to create and manage staff work shifts so that appropriate staff are available during medical-center operating hours.	5	High
E2	E2-US01	As a Patient, I want to view available appointment slots so that I can select a suitable date and time.	3	High
E2	E2-US02	As a Patient, I want to book an available appointment so that I can consult a doctor.	5	High
E2	E2-US05	As a Nurse, I want to check in arriving patients so that doctors know which patients are ready for consultation.	3	High
E2	E2-US06	As a Nurse, I want to record patient vital information so that the doctor can review the patient's condition before consultation.	3	High

E2	E2-US07	As a Doctor, I want to record consultation details so that the patient's medical history is maintained.	5	High
E2	E2-US08	As a Doctor, I want to create a prescription so that the patient can obtain the required medicines.	5	High
E2	E2-US09	As a Doctor, I want to create a laboratory test request so that the patient can complete required medical tests.	5	High
E2	E2-US11	As a Lab Assistant, I want to verify the patient's laboratory request so that I can confirm the requested test before processing it.	3	High
E3	E3-US01	As a Pharmacist, I want to add new medicines so that medicines available at the medical center are recorded in the system.	3	High
E3	E3-US02	As a Pharmacist, I want to record medicine batches so that stock can be monitored according to batch and expiry information.	5	High
E3	E3-US03	As a Pharmacist, I want to view all medicines and available quantities so that I can monitor current inventory.	3	High
E3	E3-US04	As a Pharmacist, I want to search for medicines so that I can	2	High

		quickly find medicine information.		
E3	E3-US07	As a Pharmacist, I want to update stock quantities so that inventory levels reflect actual stock availability.	3	High
E3	E3-US10	As a Pharmacist, I want to record supplier information so that the source of medicine purchases can be tracked.	3	High
E3	E3-US11	As a Pharmacist, I want to record medicine purchases from suppliers so that new stock can be added accurately.	5	High
E3	E3-US12	As a Manager, I want to receive low-stock alerts so that medicines can be reordered before they become unavailable.	3	High
E3	E3-US13	As a Manager, I want to receive medicine expiry alerts so that expired medicines and unnecessary wastage can be reduced.	3	High

E3	E3-US16	As a Pharmacist, I want to sell medicines through the system so that pharmacy transactions are properly recorded.	8	High
E4	E4-US01	As a Cashier, I want to generate an invoice from finalized medical services so that the patient can pay the correct amount.	5	High
E4	E4-US02	As a Patient, I want to view my invoices so that I know how much I need to pay.	2	Medium
E4	E4-US04	As a Cashier, I want to record patient payments so that invoices can be marked as paid.	5	High
E4	E4-US05	As a Cashier, I want to verify payments so that only valid transactions are recorded.	3	High
E4	E4-US07	As a Cashier, I want the system to generate a receipt after successful payment so that the patient receives proof of payment.	3	High
E4	E4-US11	As an Admin or Manager, I want to maintain staff salary information so that salaries can be calculated correctly.	3	High
E4	E4-US12	As a Manager, I want the system to obtain staff attendance records so that salary calculations are based on actual attendance.	5	High
E4	E4-US13	As a Manager, I want the system to calculate each employee's monthly salary so that payroll can be processed efficiently.	8	High
E4	E4-US14	As a Manager, I want to review calculated salaries before payment so that incorrect payroll calculations can be identified.	3	Medium
E4	E4-US17	As a Manager, I want the system to generate a payslip so that each employee can see details of their salary calculation.	5	High

11. Software and Hardware Requirements

11.1 Software Requirements

Category	Software / Service	Purpose
Frontend	React with Vite	Responsive role-based web interface and reusable UI components.
Backend	Node.js and Express.js	REST API, workflow logic, validation and authorization.
Database	PostgreSQL or MySQL	Relational storage for users, staff, clinical records, inventory, billing and payroll.
ORM / Data Access	Prisma ORM or equivalent	Structured database access, migrations and transaction support.
Authentication	Secure session or token-based authentication with RBAC	User authentication and role/permission enforcement.
Source Control	Git and GitHub	Version control, collaboration, issue tracking and code review.
Development IDE	Visual Studio Code	Primary coding and debugging environment.
API Testing	Postman	REST API testing and request/response validation.
UI/UX Design	Figma	Wireframes, interface design and prototype review.
Diagramming	Draw.io / PlantUML	Architecture, activity and supporting project diagrams.
Deployment	Vercel plus compatible	Prototype deployment and demonstration
	backend/database hosting	environment.
Notifications	Email/in-app notifications as implemented	Appointment/lab/status notifications within the approved MVP scope.

11.2 Hardware Requirements

Hardware	Minimum / Proposed Specification	Purpose
Development laptops/PCs	64-bit laptop/PC, Intel Core i5 / AMD Ryzen 5 class or equivalent, 8 GB RAM minimum, 20 GB free storage	Development, local testing, database tools and browser-based services.
Test mobile device	Android or iOS smartphone with a current browser	Responsive usability testing for patient and staff views.
Internet connection	Stable broadband/Wi-Fi connection	Source control, cloud database, deployment and team collaboration.
Cloud infrastructure	Free/student-tier hosted application and database resources	Shared development/demo environment without dedicated onpremises hardware.

12. Budget

Health Guard will primarily use open-source software, free-tier cloud services and existing team hardware to minimize development cost. Minor operational expenses may still arise from internet/data usage, meetings, printing and optional deployment services. The budget therefore represents an estimated academic project cost rather than commercial development pricing.

Budget Item	Basis	Estimated Cost (LKR)
React, Node.js, Express.js, Git, VS Code	Open-source / free	0
GitHub repository and collaboration	Free tier	0
PostgreSQL/MySQL database hosting	Free/student tier for prototype	0
Frontend/backend prototype hosting	Free/student tier for prototype	0
Figma, Draw.io / PlantUML, Postman	Free/student use	0
Existing laptops and mobile test devices	Existing team resources	0
Optional domain / email / cloud overage contingency	Only if required beyond free tiers	5,000

13. References

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- Atlassian. “How to write SMART goals,” Atlassian Teamwork Lab, 2023.
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- Express.js. Express web framework documentation.
- Business Architecture Assessment Report.

13. Appendix

Appendix A - Epic Ownership and Scope Boundaries

Epic	Owner	Main Process Area	Key Boundary
E1	Gawrawa G H Y - IT24100799	User & Staff Management	Patient/staff accounts, roles, shifts, attendance and leave; excludes clinical diagnosis, medicine inventory ownership and patient billing ownership.
E2	Perera M U E - IT24101266	Appointment & Clinical Workflow	Appointments, check-in, vitals, consultation, prescription and internal laboratory workflow; excludes medicine stock ownership and payment reconciliation.
E3	Bandara N W C D - IT24100264	Inventory & Procurement	Medicines, suppliers, batches, procurement, stock, expiry, pharmacy sales and stock reports; receives prescriptions but excludes clinical decision ownership.
E4	Gunasekara W L L - IT24100344	Billing & Payment Management	Patient invoices/payments/receipts/revenue and attendance-based payroll; consumes completed service and attendance data but excludes external payment gateway integration.